

Oswald Gyabaah

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PROFESSIONAL SUMMARY

Fullstack engineer with 6 years building and shipping production web, mobile, and AI-powered systems end-to-end. Founding CTO of [MyHealthCop](#) - designed and built 3 production apps (React admin + Flutter iOS/Android) on AWS EKS with 1M+ daily background jobs and a 90% peak-latency cut through PostgreSQL tuning and Redis caching. Founder of [erifoundry.org](#), an EdTech research foundry prototyping LLM applications in human-centered learning. Daily work in TypeScript, Next.js, Node.js, PostgreSQL, Redis, and Docker, plus production LLM agents using OpenAI and Anthropic SDKs with RAG, embeddings, and tool-use patterns.

CORE COMPETENCIES

TypeScript & Node.js

Next.js & React

LLM Integration (OpenAI, Anthropic)

RAG & Embeddings

PostgreSQL & Redis

REST APIs & Service Layer

Background Jobs & Workers

AI Product Engineering

WORK EXPERIENCE

Guaranty Trust Bank Ghana LTD

May 2025 - Present

Core Banking Services Lead - Accra, Ghana

- Built a bank-wide OTP service in **Next.js + Django**, integrating SMS and email gateways to secure core banking access for 600+ internal users, with structured logging to Elasticsearch via Filebeat.
- Built a national ID verification service in **Next.js + Django**, processing and storing ID documents in self-hosted MinIO with automated verification and customer record updates - a file-upload, document-heavy workflow in production.
- Architected integrations between Finacle core banking and third-party systems including transaction portals and internal tools, using **REST APIs** and service-layer patterns.
- Own the on-prem Finacle deployment on Tanzu Kubernetes Grid, managing 800+ production pods to sustain 1.5M+ daily transactions; lead L2 incident response correlating logs across services via the Elastic Stack.
- Extended observability with Finacle-specific Grafana dashboards (CPU/memory, transaction volumes, response times, exceptions) plus threshold-based alerting for proactive scaling.

Guaranty Trust Bank Ghana LTD

Apr 2024 - Present

DevOps Engineering Lead - Accra, Ghana

- Founded the bank's DevOps team of 4 engineers; replaced manual code handoffs with enforced Git workflows, containerized Kubernetes deployments, and automated release pipelines.
- Implemented CI/CD automation that cut deployment time ~90% and increased release velocity ~5x; introduced GitOps with FluxCD; integrated Veracode and Trivy for automated security scanning.
- Architected HA Kubernetes clusters across on-prem environments running internal applications handling 3M+ daily requests at 99.97% uptime; implemented auto-scaling and self-healing.
- Automated 95% of infrastructure configuration management with Ansible, Python, and Bash, turning multi-day manual processes into reproducible deployments in minutes.
- Unified observability with **Sentry**, Prometheus, Grafana, and the Elastic Stack for centralized logging, tracing, and real-time alerting - reducing MTTR on production incidents.

Chief Technical Officer (Founding) - Accra, Ghana

- Architected [myhealthcop.com](#) end-to-end and shipped 3 production apps: a patient Flutter mobile app (iOS + Android), a Flutter mobile app for health professionals, and a **React admin web dashboard** - fullstack ownership from requirements through delivery.
- Designed secure, cost-efficient AWS infrastructure on EKS and ECS alongside Lambda for event-driven workloads, with autoscaling, load balancing, and inter-service communication.
- Built a high-throughput background processing system handling 1M+ daily jobs across payments, messaging, and notifications using SQS/SNS, with SMS, email, and payment provider integrations including multi-provider failover.
- Led performance optimizations that reduced peak request latency by **90%** through Redis caching, query tuning, and PostgreSQL indexing.
- Automated application delivery and infrastructure provisioning via CI/CD pipelines on GitHub Actions and Terraform - repeatable, auditable, reversible.
- Designed a disaster recovery strategy mirroring the AWS production environment on GCP (GKE, Cloud SQL, Cloud Storage) with secure VPCs, IAM policies, and dual-environment CI/CD for rapid failover.
- Led a 12-person cross-functional team (frontend, backend, mobile, product, design, QA) through requirements gathering, sprint planning, stakeholder management, and budgeting under Agile/Scrum.

PROJECTS

erifoundry.org - Education Research and Innovation Foundry EdTech AI

Founder & Research Lead. Leading design of AI-driven research initiatives in adaptive learning, cognitive modeling, and educational analytics. Coordinating cross-disciplinary research on ML and software tooling for education. Prototyping applications of LLMs in human-centered learning systems.

Stack: LLM APIs (OpenAI, Anthropic), RAG, embeddings, Python, Next.js

AI Agents for Infrastructure Operations LLM in Production

Building specialized AI agents for QA automation, code review, and sysadmin operations, running against live Linux production servers with real workloads. Prototyping agent-driven SaaS tools using RAG, tool-use, and multi-agent orchestration patterns. Built AWS Lambda functions triggered by EventBridge to deliver notifications on codebase events.

Stack: Anthropic API, OpenAI API, RAG, tool-use, AWS Lambda, EventBridge, DynamoDB, Python

EDUCATION

BSc Computer Engineering - Ashesi University

Sep 2016 - May 2020

Berekuso, Ghana

CERTIFICATIONS

AI Engineering Specialization - ByteByteAI	Mar 2026
Oracle Certified DevOps Professional - Oracle	Aug 2025
Oracle Certified Architect Associate - Oracle	Aug 2025
AWS Certified Cloud Practitioner - AWS	May 2024
Django Web Framework - Coursera	Dec 2022

SKILLS

Frontend & Mobile: TypeScript, JavaScript, React.js, Next.js, Tailwind CSS, Flutter

Backend: Node.js (Express, service-layer APIs), Python (Django, FastAPI, Celery), REST APIs, GraphQL, WebSockets

AI & LLM: OpenAI, Anthropic, Gemini API integration; RAG, embeddings, prompt design, structured outputs; AI agent development, multi-agent orchestration, tool-use patterns

Databases: PostgreSQL, Redis, MySQL, MongoDB, Firebase/Firestore, DynamoDB, Convex, SQLite

Testing: Vitest, Jest, React Testing Library, Cypress, Playwright E2E, pytest

Infrastructure & DevOps: Docker, Kubernetes (EKS, GKE, TKG, RKE2), Terraform, Ansible, GitHub Actions, ArgoCD, FluxCD, AWS, GCP, Azure, OCI

Observability: Sentry, Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana, Filebeat)

Messaging & Queues: AWS SQS/SNS, AWS EventBridge, RabbitMQ, Redis queues, Kafka